Process Characterization Report

A-Mab Drug Substance — Low-pH Viral Inactivation (Step 6) — PCR-006

Process Development / Manufacturing Science & Technology (MSAT)

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Abbreviations

AMV, analytical method validation; ANOVA, analysis of variance; CCD, central composite design; CPP, critical process parameter; CQA, critical quality attribute; DCS, distributed control system; DoE, design of experiments; DS, drug substance; GPP, general process parameter; HMW, high molecular weight; icIEF, imaged capillary isoelectric focusing; IgG1, an immunoglobulin G subclass; LRF, log reduction factor; MSAT, Manufacturing Science and Technology; NOR, normal operating range; PAR, proven acceptable range; PCP, process characterization plan; PCR, process characterization report; PD, Process Development; QA, Quality Assurance; RSM, response surface methodology; SDM, scale-down model; SEC-HPLC, size-exclusion high-performance liquid chromatography; SOP, standard operating procedure; TCID50, fifty percent tissue culture infective dose; WC-CPP, well controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

Executive summary

The low-pH viral inactivation step holds the Protein A eluate at acid pH for a defined time before it is neutralised and passed to cation exchange chromatography. Acid disrupts the lipid envelope of an enveloped virus and denatures the glycoproteins it needs to enter a cell, so the hold inactivates retrovirus-like particles without a separation. The step is the first of the three unit operations credited with virus clearance in the A-Mab drug substance process, and it carries the largest single contribution to the XMuLV claim at 7.10 $\log_{10}$. This report records the characterization of the step executed against the protocol PCP-006 on a qualified small-scale spiking model, for a commercial process operated at 15,000 L.

4 parameters were carried into the study. 3 of them were studied in a multivariate design and 1 was assessed one at a time. Inactivation pH was classified as a critical process parameter, hold time and temperature as well controlled critical process parameters, and A-Mab concentration as a general process parameter (§9). The step sets one critical quality attribute, the cumulative clearance of XMuLV, and it acts on two attributes that other steps control, aggregate and acidic charge variants (§2.2).

Screening over 11 runs identified inactivation pH, hold time and temperature as active on virus inactivation, and hold time and temperature as active on aggregate (§5.2). The response-surface designs of 18 runs refined those relationships into the predictive models used for every claim in this report. The model for XMuLV log reduction returned R^2 of 0.896 and the model for aggregate R^2 of 0.997, and neither showed significant lack of fit against the centre-point pure error (§5.3).

The characterized region does not meet every criterion at every corner. 79.9% of an evenly spaced grid over the characterized ranges satisfies all three criteria at once, and aggregate is the response that rejects the rest (§6). At the worst corner of the characterized region the model predicts an XMuLV log reduction of 4.91 $\log_{10}$ against a required step contribution of 4.93 $\log_{10}$, so an edge of failure lies inside the knowledge space. Every corner of the normal operating ranges meets both criteria with margin (§7).

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches with all parameters inside their normal operating ranges. The cumulative XMuLV clearance returned a one-sided Cpk of 1.53, with a mean 2.81 $\log_{10}$ above the requirement (§8). That figure is a claim about the train and not about this step alone, since anion exchange and virus filtration contribute 11.77 $\log_{10}$ of the total (§10).

2 deviations were recorded during execution. Both were investigated, both were bounded against the fitted models or the analytical method, and both were retained with the affected data included in the analysis (§13). The parameter classification, the

proven acceptable ranges and the clearance contribution reported here roll up into PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody produced by fed-batch mammalian cell culture and purified on a platform train of six downstream unit operations. The train runs Protein A capture, low-pH viral inactivation, cation exchange chromatography, anion exchange chromatography, small-virus retentive filtration and ultrafiltration with diafiltration. This report covers Step 6 of that train.

The step receives the Protein A eluate, which is already at acid pH from the elution buffer. The pool is titrated to the inactivation pH, held for the specified time at controlled temperature, and then neutralised for the cation exchange load. Inactivation of an enveloped virus at low pH follows first-order kinetics whose rate constant rises as the pH falls and as the temperature rises, so the log reduction achieved is set by how acidic the hold is, how long it lasts and how warm it is. No separation occurs and no impurity is removed by design.

The same conditions act on the antibody. At acid pH the molecule carries a large net positive charge and the second constant domain of the heavy chain is partially unfolded, which exposes hydrophobic surface and allows molecules to associate. Aggregate therefore forms during the hold, at a rate that increases with time and temperature. Time at low pH can also add acidic species through acid-catalysed reactions at labile residues, including succinimide formation and hydrolysis.

The step forms no critical quality attribute other than the virus clearance it delivers. Aggregate formed during the hold is carried into cation exchange, which is the step that removes it (PCR-007). Precipitation of host cell protein is often observed during a low-pH hold, but the extent is not predictable from the parameters studied here, and no host cell protein clearance is claimed for this step. The host cell protein level of the Protein A pool is therefore treated as the input to cation exchange without credit for the hold.

1.2 Objectives

The study had five objectives.

1. Quantify the relationship between the hold parameters and the responses the step governs, over ranges wider than the routine control ranges.
2. Define the operating region within which the governed attributes stay inside their acceptance criteria.

3. Establish a proven acceptable range for each parameter against each governed attribute, at fixed conditions and with the remaining parameters varying inside their normal operating ranges.
4. Classify each parameter according to its demonstrated effect and the risk of operating outside the acceptable region.
5. Justify the ranges to be confirmed in Stage 2 and state this step's contribution to the control strategy.

1.3 Regulatory and scientific basis

The study follows the lifecycle approach to process validation, in which process design studies of this kind constitute Stage 1 (U.S. Food and Drug Administration 2011). The enhanced approach of ICH Q8, the quality risk management principles of ICH Q9 and the development principles of ICH Q11 govern how the parameters were prioritized, studied and classified (International Council for Harmonisation 2009, 2023b, 2012). Criticality is treated as a continuum. A parameter with a demonstrated effect on a critical quality attribute is therefore classified by the size of that effect together with the reliability with which the parameter is controlled.

Virus clearance is evaluated as a modular claim under ICH Q5A(R2) (International Council for Harmonisation 2023a). Each contributing step is studied independently on a small-scale spiking model, and the log reduction factors of steps that clear virus by different mechanisms are added to give the cumulative claim. The process model, the quality attribute framework and the risk methodology follow the A-Mab case study (CMC Biotech Working Group 2009).

The documents that frame or consume this report are listed in Table 1.1. PCP-006 is the approved protocol this report executes, RA-001 set the scope of the study, and PCMR-001 consolidates the outcome with the other unit operations.

Table 1.1: Related documents in the A-Mab characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-006	Process Characterization Plan (Low-pH Viral Inactivation (Step 6))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The low-pH hold has been operated at Novacyte on the humanized IgG1 platform through three previous products, X-Mab, Y-Mab and Z-Mab. The hold conditions have remained essentially unchanged across those products and through A-Mab development, so the platform experience constitutes prior product knowledge for this step. That knowledge establishes the mechanism and the direction of every effect studied here. It does not establish the magnitudes for A-Mab, because the rate at which a given antibody aggregates at acid pH depends on the stability of its own second constant domain.

Three expectations were carried into the study from that experience. Inactivation pH was expected to be the parameter the virus inactivation depends on most steeply, because the rate constant of envelope disruption and glycoprotein denaturation rises sharply as the pH falls across this range. Hold time was expected to raise the log reduction, because the hold integrates the inactivation rate, and to raise aggregate for the same reason, because it gives the partially unfolded antibody longer to associate. Temperature was expected to act in the same direction on both responses, since it accelerates the denaturation that inactivates the virus and the association that forms aggregate.

A-Mab concentration was expected to act on aggregation and not on inactivation. Association is a bimolecular event and proceeds faster in a more concentrated solution, while the acid chemistry that inactivates the virus does not depend on how much antibody is present. Concentration is a property of the Protein A pool and is not set at the hold, which is why it was assessed one parameter at a time.

Prior experience also set the lower bound of the pH range studied. Antibody precipitation has been observed on this platform below the low edge of the range in Table 4.1, and the range was bounded above that point so that precipitation did not confound the study.

2.2 Quality attributes in scope

The attributes this step governs are listed in Table 2.1 with their acceptance criteria and their criticality. Criticality was assigned in RA-001 by the impact and uncertainty method of the A-Mab case study, in which an attribute scored high or very high is treated as critical (CMC Biotech Working Group 2009).

Table 2.1: Quality attributes governed by the low-pH viral inactivation step.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4

The step sets one of these attributes. Clearance of XMuLV is a viral safety attribute of very high criticality, and its acceptance criterion is cumulative across the train rather than assigned to any single step. The other two attributes are formed in the production bioreactor and are reported here because the hold moves them. Aggregate rises during the hold and is reduced at cation exchange. Acidic charge variants can rise during the hold and are of very low criticality.

Because the XMuLV criterion is cumulative, the criterion this step was judged against is the step contribution required once the credited clearance of the other steps is taken into account. That figure is 4.93 $\log_{10}$, obtained by subtracting the 11.77 $\log_{10}$ credited to anion exchange and virus filtration from the cumulative requirement. It is used as the acceptance criterion throughout §7 and is not a claim that the step must deliver the whole of the drug substance requirement.

Aggregate was judged against an in-process criterion of 2.17 %. That criterion is the level this step may hand on and still leave the drug substance inside its limit after the downstream train has cleared aggregate, divided by an assurance margin. Judging an intermediate against the drug substance specification would credit the hold with clearance it does not perform, and would report the step as acceptable at settings the polishing step could not recover.

2.3 Risk-based prioritization and parameter selection

RA-001 ranked every parameter of the step by its potential impact on a quality attribute and by its potential to interact with another parameter, and used the combined rank to assign a study type. Three parameters reached the threshold for multivariate study. Inactivation pH, hold time and temperature all act on the inactivation kinetics and on the aggregation that proceeds in parallel with it, so their effects could not be separated one at a time. A-Mab concentration fell below that threshold and was assigned to univariate assessment, since it does not enter the inactivation chemistry and is bounded by the Protein A pool specification.

The ranges studied were widened beyond the routine control ranges, which is what makes this a characterization study. Table 4.1 gives both. The characterization ranges cover the excursions a manufacturing suite can plausibly produce and extend far enough to locate an edge of failure where one exists. The pH range was bounded below by the precipitation limit described in §2.1 and above at the point where prior knowledge indicated that inactivation within the shortest hold could no longer be assured.

3 Materials and methods

3.1 Scale-down model and its qualification

The studies were executed on a scale-down model of the commercial hold vessel, operated under SOP-2009 and qualified under SOP-1001. The model reproduces the commercial operation by holding the scale-independent variables at their commercial values. Inactivation pH, hold time, temperature and A-Mab concentration are scale-independent, so the model was operated at the same values the commercial vessel is controlled to. Mixing was set to give the same time to homogeneity after acid addition as the commercial vessel, so that the titration transient is comparable at the two scales.

Qualification compared the model with at-scale data on the inputs and the outputs of the step. The input attributes compared were the Protein A pool concentration, its aggregate content and its charge-variant distribution. The output attributes compared were the aggregate content and charge-variant distribution of the neutralised pool. No difference was found that would change the conclusions drawn on the model, and the model was accepted as representative of the commercial step for the purposes of this characterization.

Virus clearance cannot be measured at commercial scale, because live virus may not be introduced into a manufacturing suite. The log reduction factors reported here were therefore measured on a qualified small-scale spiking model, which is the approach ICH Q5A(R2) requires and which the modular clearance claim rests on (International Council for Harmonisation 2023a). The spiking model uses the same vessel geometry, the same acid and base additions and the same hold control as the scale-down model, with XMuLV spiked into the load at a titre high enough to allow a reduction of the size claimed to be measured. Spiked material was not returned to any process stream.

The reproducibility of the model is reported in §5.1. The replicate centre points of each design supply the pure-error term against which lack of fit was tested, so the same runs that qualify the model quantitatively also bound the models fitted to it.

3.2 Operation

The step was operated as the commercial process will be operated. The Protein A eluate was adjusted to the target inactivation pH with acid, the pH was confirmed before the hold started, and the hold was timed from that confirmation. Temperature was controlled for the duration of the hold. At the end of the hold the pool was titrated to the cation exchange load pH with base, and the neutralised pool was filtered before the next step.

Two inputs were controlled to platform specification and were not characterized. The acid and base solutions were prepared and released under the buffer preparation procedure, and the Protein A pool was released against its own in-process controls. Materials controlled this way are identity-controlled and quality-controlled inputs to the study. They were not varied.

3.3 Analytical methods

Table 3.1 lists the controlled documents that governed the study and the validated methods that produced its data. XMuLV titre was determined by infectivity assay under AMV-3017, aggregate by size-exclusion chromatography under AMV-3011, and charge variants by imaged capillary isoelectric focusing under AMV-3013.

Table 3.1: Controlled documents and validated methods governing the study.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2009	Operation of the Low-pH Viral Inactivation Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

The log reduction factor is the difference between the virus titre of the spiked load and the titre of the held pool, each expressed as $\log_{10}$ TCID50 per millilitre and corrected for the volumes involved. It is therefore a difference of two assay results, and it carries the uncertainty of both. The reported precision and quantitation limit of the infectivity assay are given in Appendix D. A held pool whose titre falls to the quantitation limit yields a log reduction factor that is a lower bound rather than a point estimate, which is the convention used throughout this report and the reason the clearance claim is conservative.

3.4 Sampling plan

Each run was sampled at the start and at the end of the hold. The load sample established the input titre, the input aggregate content and the input charge-variant distribution. The end-of-hold sample was taken immediately before neutralisation and established the output values of the same three attributes. Samples for infectivity assay were held at the assay laboratory's storage condition and tested within the qualified interval.

Centre points were replicated so that reproducibility could be estimated from the design itself. The screening design carried 3 centre-point replicates among its 11 runs, and the response-surface design carried 4 among its 18 runs. The centre condition was replicated in preference to a corner, which keeps the pure-error estimate in the middle of the region studied, where every model is used.

3.5 Statistical methods

Factors were coded so that the coded levels −1, 0 and +1 correspond to the low edge, the midpoint and the high edge of each characterization range. Coding puts the estimated effects of parameters with different units on a common scale, and it makes the coefficient of a term the change in the response over half the characterization range. Analyses were performed under SOP-1002 at a significance level of 0.05.

The screening designs were analysed by ordinary least squares with all main effects and all two-factor interactions. An effect is reported as twice the coded coefficient, which is the change in the response between the low and the high edge of the range. The screening model identifies which parameters are active. It does not describe curvature, and its estimates are refined by the response-surface model, which is the predictive model used for every range and every prediction in this report.

The response-surface designs were analysed with the full quadratic model, containing the main effects, the two-factor interactions and the pure quadratic terms. Model adequacy was judged on four grounds. R^2 states how much of the variation the model reproduces, adjusted R^2 penalises terms that do not earn their degrees of freedom, predicted R^2 from the predicted residual error sum of squares states how the model performs on a run it did not see, and the analysis of variance partitions the residual into lack of fit and pure error. A model whose lack-of-fit F is not significant against the replicated centre points is accepted as adequate over the region studied. A parameter with no significant term in the model is reported as one over which the response is robust across the range studied.

Proven acceptable ranges were computed from the fitted response-surface models by two analyses, described in §7. Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches with every parameter drawn inside its normal operating range, and is reported as a one-sided Cpk against the criterion that applies to each attribute.

4 Study design

4.1 Factors, ranges and the knowledge space

Table 4.1 gives the four parameters, their set-points, their normal operating ranges, the ranges characterized in this study and the classification each received from the results. The normal operating range is the range the commercial process is controlled within. The characterization range is the knowledge space, and it is wider by design, so that the behaviour of the step is known outside the range it will be operated in.

Table 4.1: Parameters, ranges and final classification for the low-pH viral inactivation step.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Inactivation pH	pH	3.5	3.4–3.6	3.2–4	CPP	multivariate
Hold time	min	90	60–120	60–180	WC-CPP	multivariate
Temperature	°C	21	19–23	15–25	WC-CPP	multivariate
A-Mab concentration	g/L	20	5–35	5–35	GPP	univariate

The inactivation pH range is the narrowest relative to its set-point, at 3.2 to 4 around a set-point of 3.5. That follows from the chemistry and not from the control system. The inactivation rate depends steeply on pH, and the antibody precipitates below the low edge, so the useful range is bounded on both sides. The hold time range extends from 60 to 180 minutes, twice the width of the normal operating range, and the temperature range from 15 to 25 °C, which brackets what a manufacturing suite delivers at ambient control.

The coded letters used in the effect and coefficient tables are given in Table 4.2. All three multivariate parameters were studied in both designs.

Table 4.2: Coded factor legend for the screening and response-surface designs.

Code	Factor	Unit	Studied in
A	Inactivation pH	pH	screening + RSM
B	Hold time	min	screening + RSM
C	Temperature	°C	screening + RSM

4.2 Screening design

Screening used a two-level full factorial in the three multivariate parameters, augmented with centre points. The design ran 11 runs, of which 8 were factorial and 3 were replicates of the centre condition. A full factorial in three factors estimates every main effect and every two-factor interaction without aliasing, so no assumption about negligible interactions was needed. The runs were executed in randomised order. The design matrix and the measured responses are given in Appendix A.

Three responses were measured on every run. Aggregate and acidic charge variants were measured on all runs by the methods in §3.3. Virus titre was measured on the spiked counterpart of each run.

4.3 Response-surface design

The response-surface design was a face-centred central composite design in the same three parameters, run over 18 runs, of which 4 were centre-point replicates. A face-centred design places its axial points on the faces of the coded cube, so it adds curvature without requiring any parameter to be set outside the characterization range. That property matters here, because the low edge of the pH range is bounded by antibody precipitation and an axial point beyond it could not have been executed. The design matrix and the measured responses are given in Appendix B.

All three screening parameters were carried into the response-surface design. Screening had shown that each of them is active on at least one response, so none could be dropped without losing a term the predictive model needs.

4.4 Univariate assessment

A-Mab concentration was assessed one parameter at a time across the range in Table 4.1, with pH, hold time and temperature at their set-points. The parameter was studied this way for two reasons. It does not enter the inactivation chemistry, so no interaction with the parameters that do was expected, and it is set by the Protein A pool, so a multivariate design would have studied a variable the operator does not choose.

The assessment covers the concentration range the Protein A pool delivers. Outside that range the assessment says nothing, and a change to the capture step that moved the pool concentration beyond it would require the assessment to be repeated. The outcome is reported with the parameter classification in §9.

5 Results

5.1 Centre-point performance and reproducibility

The replicated centre condition sits at the midpoint of each characterization range, at pH 3.6, a hold of 120 minutes and 20 °C. Table 5.1 gives the mean, the standard deviation and the coefficient of variation of each response across the 4 replicates of the response-surface design, and Table 5.2 gives the same for the 3 replicates of the screening design. These runs are the pure-error term against which lack of fit is tested in §5.3.

Table 5.1: Centre-point reproducibility, response-surface design.

Response	n	Mean	SD	%CV
Aggregates (HMW, %)	4	2.01	0.0169	0.845
XMuLV LRF ($\log_{10}$)	4	6.71	0.381	5.67
Acidic variants	4	20.7	0	0

Table 5.2: Centre-point reproducibility, screening design.

Response	n	Mean	SD	%CV
Aggregates (HMW, %)	3	2.03	0.0324	1.6
XMuLV LRF ($\log_{10}$)	3	7.21	0.253	3.51
Acidic variants	3	20.7	0	0

The least precise response is the XMuLV log reduction factor, at 5.7 % in the response-surface design and 3.5 % in the screening design. In absolute terms the standard deviation of the replicates is 0.38 $\log_{10}$. That figure exceeds the reported precision of the infectivity assay by a wide margin (Appendix D), which indicates that the spread comes from the run and not from the measurement. An infectivity assay reads a biological endpoint on a dilution series, and the inactivation itself proceeds on a spiked load whose starting titre is not identical between runs.

Aggregate reproduces to 0.84 % in the response-surface design, so a change of 0.017 % aggregate is at the level of the noise. Acidic charge variants returned the same value at every replicate, to the precision the method reports. The consequence is a pure-error term of zero for that response, which is taken up in §5.3.

5.2 Factor effects from the screening design

Table 5.3 summarises the screening models. The two responses that vary between replicates are described well by a model containing only main effects and two-factor interactions. The third is treated in §5.3.

Table 5.3: Screening model summary across the three responses.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Aggregates (HMW, %)	11	0.992	0.98	0.828	82	0.000386	0.0333
XMuLV LRF ($\log_{10}$)	11	0.982	0.956	0.819	37.2	0.00182	0.265
Acidic variants	11	1	1	1	3.93e+26	1.73e-53	1.16e-14

The effects on virus inactivation are given in Table 5.4. Inactivation pH is the largest, at -2.08 $\log_{10}$ across the characterization range ($p < 0.001$), and it acts in the negative direction. Raising the pH from the low edge to the high edge of the range costs that much clearance. Hold time is next, at +1.72 $\log_{10}$ ($p < 0.001$), and temperature third, at +0.64 $\log_{10}$ (p 0.027). No interaction reached significance. The largest of them, between hold time and temperature, is +0.24 $\log_{10}$ with p 0.272.

Table 5.4: Screening effect estimates for XMuLV log reduction.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	-2.08	-1.04	0.0936	-11.1	0.000376	***
B	1.72	0.862	0.0936	9.2	0.000774	***
C	0.636	0.318	0.0936	3.4	0.0274	*
BC	0.238	0.119	0.0936	1.27	0.272	
AB	0.203	0.101	0.0936	1.08	0.34	
AC	-0.194	-0.097	0.0936	-1.04	0.359	

The effects on aggregate are given in Table 5.5 and rank differently. Hold time is the largest at +0.471 % ($p < 0.001$), temperature second at +0.221 % ($p < 0.001$), and inactivation pH did not reach significance at +0.022 % (p 0.402). The parameter that dominates virus inactivation therefore has no demonstrated effect on aggregate over the range studied, and the two parameters that dominate aggregate are the two that come second and third on inactivation.

Table 5.5: Screening effect estimates for aggregate.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	0.471	0.236	0.0118	20	3.69e-05	***
C	0.221	0.11	0.0118	9.37	0.000723	***

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AC	0.035	0.0175	0.0118	1.48	0.212	
AB	0.0285	0.0142	0.0118	1.21	0.294	
A	0.0221	0.011	0.0118	0.937	0.402	
BC	0.0107	0.00535	0.0118	0.453	0.674	

Acidic charge variants responded to hold time alone, by +0.40 % across the characterization range. No other term returned an estimate distinguishable from zero. The full table is in Appendix C. The change is small against the criterion of 40 % and against the level the production bioreactor delivers, and it does not constrain the step.

The screening design identifies which parameters are active and estimates their effects on the assumption that the response is linear between the two edges of each range. It cannot describe curvature, and the estimates above are superseded by the response-surface models in §5.3 wherever the two disagree.

5.3 Response-surface models

Table 5.6 summarises the three response-surface models.

Table 5.6: Response-surface model summary across the three responses.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Aggregates (HMW, %)	18	0.997	0.993	0.961	261	6.57e-09	0.0169
XMuLV LRF ($\log_{10}$)	18	0.896	0.779	0.579	7.67	0.00433	0.367
Acidic variants	18	1	1	1	5.99e+26	2.43e-106	8.61e-15

The aggregate model reproduces the data closely, with R^2 of 0.997, adjusted R^2 of 0.993 and predicted R^2 of 0.961. The closeness of the three statistics indicates that the model is not carrying terms that fail to earn their degrees of freedom, and that it predicts a run it did not see about as well as it fits the runs it did. Its coefficients are given in Table 5.7. Hold time and temperature are the significant terms, at +0.236 % and +0.107 % per coded unit. No quadratic term reached significance, so the surface is close to a plane over the region studied.

Table 5.7: Response-surface coefficients for aggregate.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	2.01	0.00667	301	1.64e-17	***
A	-0.00244	0.00536	-0.455	0.661	

Term	Coef.	Std. err.	t	p-value	Sig.
B	0.236	0.00536	44.1	7.76e-11	***
C	0.107	0.00536	20	4.01e-08	***
AB	-0.00133	0.00599	-0.221	0.83	
AC	0.0133	0.00599	2.21	0.0578	
BC	0.00837	0.00599	1.4	0.2	
A^2	-0.00247	0.0103	-0.24	0.816	
B^2	0.000627	0.0103	0.0609	0.953	
C^2	0.0125	0.0103	1.21	0.261	

The XMuLV model is given in Table 5.8. Inactivation pH and hold time are the significant main effects, at -0.612 and +0.605 $\log_{10}$ per coded unit, and temperature follows at +0.286 $\log_{10}$ per coded unit (p 0.039). The pure quadratic term in pH is -0.513 $\log_{10}$ with p 0.050, which sits at the significance level and not inside it. The term is retained in the model because the curvature it describes is expected from the chemistry and because dropping it would flatten the surface where the clearance margin is smallest.

Table 5.8: Response-surface coefficients for XMuLV log reduction.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	6.91	0.144	47.9	3.99e-11	***
A	-0.612	0.116	-5.28	0.000747	***
B	0.605	0.116	5.22	0.000806	***
C	0.286	0.116	2.46	0.0391	*
AB	0.162	0.13	1.25	0.246	
AC	0.0691	0.13	0.533	0.608	
BC	-0.105	0.13	-0.812	0.44	
A^2	-0.513	0.223	-2.31	0.0501	
B^2	0.202	0.223	0.907	0.391	
C^2	0.152	0.223	0.684	0.513	

The XMuLV model is less precise than the aggregate model, and the difference matters for how it may be used. R^2 is 0.896, adjusted R^2 is 0.779 and predicted R^2 is 0.579. The gap between the first and the last is the model's own statement of how much less well it predicts a new run than it fits the existing ones, and it follows from the reproducibility reported in §5.1. The model is suitable for predicting mean levels of clearance over the ranges studied. It is not a statement about the clearance a single batch will achieve, and every claim built on it in §6 and §7 is either a mean prediction or a predictive interval that carries this residual forward.

Table 5.9 and Table 5.10 partition the residual of each model into lack of fit and pure error. Neither partition gives grounds to reject the model. The lack-of-fit F for aggregate is 1.00 with p 0.535, and for XMuLV 0.88 with p 0.580. In both cases the variation the model fails to explain is of the same size as the variation between replicates of the same condition, which is the definition of an adequate fit over the region studied.

Table 5.9: Analysis of variance with the lack-of-fit partition, aggregate.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	0.676	9	0.0751	261	6.57e-09
Residual	0.0023	8	0.000287	nan	nan
Lack of fit	0.00144	5	0.000287	1	0.535
Pure error	0.000861	3	0.000287	nan	nan

Table 5.10: Analysis of variance with the lack-of-fit partition, XMuLV log reduction.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	9.27	9	1.03	7.67	0.00433
Residual	1.07	8	0.134	nan	nan
Lack of fit	0.64	5	0.128	0.884	0.58
Pure error	0.435	3	0.145	nan	nan

The acidic charge-variant model requires a different statement. The replicates returned the same value to the precision the method reports, so the pure-error sum of squares is zero, the lack-of-fit test cannot be evaluated, and the F statistic and its p-value in Table 5.6 carry no information. The response over the region studied is a linear function of hold time and of nothing else, rising by 0.40 % from the shortest hold to the longest. The result is reported here as a bounded observation. No predictive model is claimed for this response, and the bound is what §7 uses.

The response surfaces of all three responses over inactivation pH and hold time are shown in Figure 5.1, with temperature held at the centre of its range. The aggregate panel shows a surface that rises with hold time and is flat in pH. The XMuLV panel shows the opposite gradient in pH and the same gradient in hold time, so the two attributes are traded against each other along the hold-time axis and are independent along the pH axis.

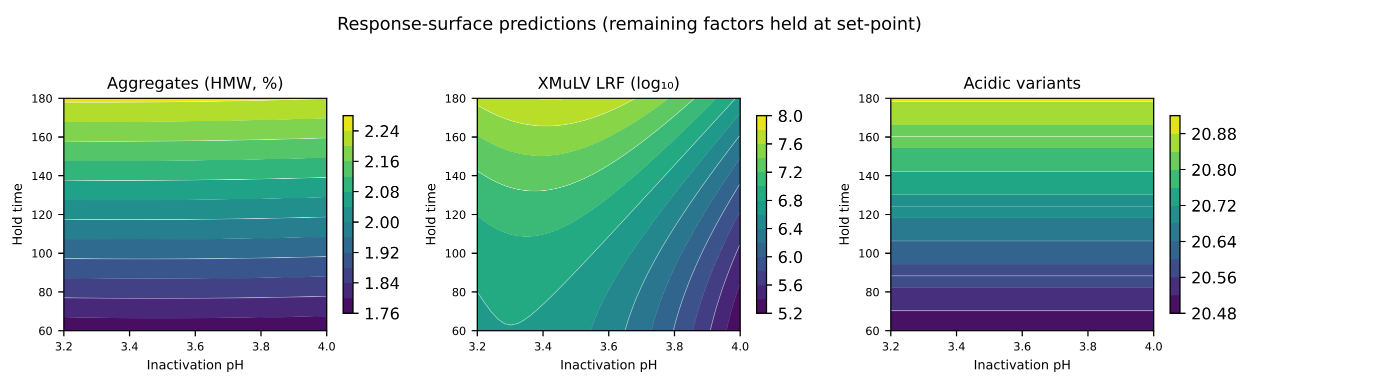

Figure 5.1: Response-surface predictions over inactivation pH and hold time, with temperature at the centre of its characterization range.

Figure 5.2 and Figure 5.3 give the residual diagnostics of the two models that carry criteria. The residuals scatter around zero without a pattern against the predicted value, the normal quantile plots are close to linear, and the predicted values track the measured values along the identity line. Nothing in either set of plots indicates a missing term or a variance that changes across the region.

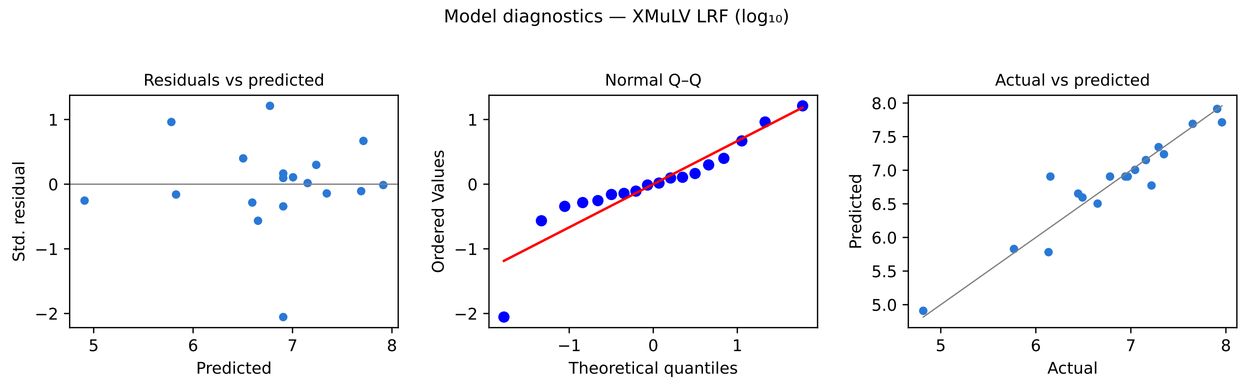

Figure 5.2: Residual diagnostics for the XMuLV log reduction response-surface model.

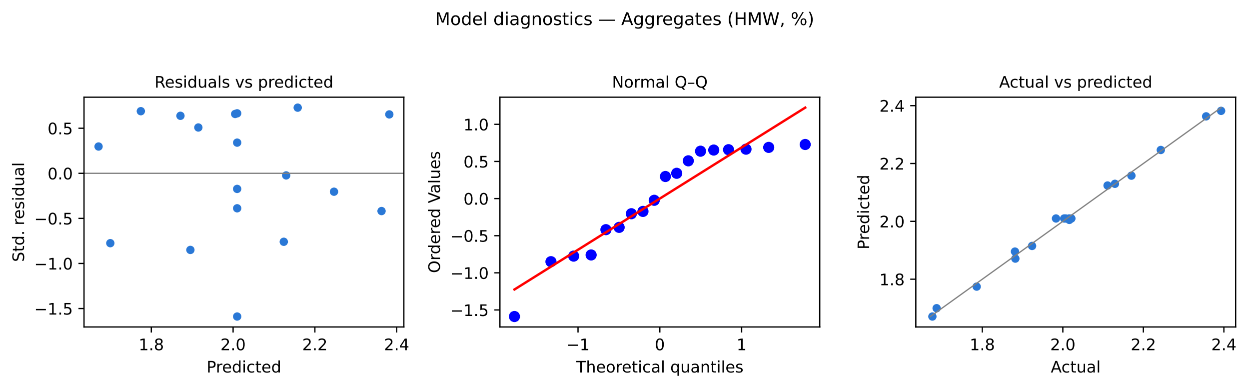

Figure 5.3: Residual diagnostics for the aggregate response-surface model.

5.4 Mechanistic interpretation

The surfaces have the shape the chemistry of the step predicts. Inactivation of an enveloped virus at low pH is a first-order process whose rate constant rises as the pH falls, because acid disrupts the lipid envelope and denatures the fusion glycoproteins the virus needs to enter a cell. The log reduction is the rate constant multiplied by the hold time, so a lower pH and a longer hold both raise it, and the model returns coefficients of the same size and opposite sign for the two (Table 5.8). Temperature raises the rate constant by accelerating the same denaturation, and its coefficient is smaller because the range studied spans only ten degrees around ambient.

The negative curvature in pH says that the gain in clearance from lowering the pH falls away toward the acid edge of the range. Two readings are available for it. Either the inactivation is no longer limited by the same step at the acid edge, or the measured reduction is bounded above by the reportable range of the infectivity assay, which the cytotoxicity described in §13.2 raised for these samples. The quadratic term is at the boundary of significance, so the model does not separate the two readings, and no conclusion is drawn from the shape beyond the one the surface supports. The clearance is claimed as a minimum in §10 for this reason.

Aggregate behaves as a bimolecular association of partially unfolded molecules. At acid pH the antibody carries a large net positive charge and the second constant domain of its heavy chain is partially unfolded, which exposes hydrophobic surface. Association then proceeds for as long as the molecules are held in that state and faster when they are warmer, which is exactly the pair of significant coefficients the model returns. The absence of a pH term is consistent with the same picture over this narrow range. Across 3.2 to 4 the molecule is already in the partially unfolded state at every setting, so moving the pH within the range changes how fast the virus is inactivated without changing how fast the antibody associates.

That separation is what makes the step operable. The parameter that governs virus inactivation does not govern aggregate, so the clearance can be assured by pH while the aggregate burden handed to cation exchange is bounded by hold time and temperature. The two constraints meet only through hold time, which raises both the clearance and the aggregate, and the operating region in §6 is the region where the hold is long enough for the first and short enough for the second.

Acidic charge variants rise with hold time through acid-catalysed reactions at labile residues, including succinimide formation and hydrolysis. Over a hold of the length studied the change is small against the level the production bioreactor sets, and the attribute is of very low criticality. The step is therefore not a control point for charge variants, and it is not treated as one in §10.

6 Design space

The design space of this step is the region of inactivation pH, hold time and temperature over which the fitted models predict that every governed attribute stays inside its criterion. It is a subset of the characterized region, and for this step it is a proper subset. Evaluating the three models on an evenly spaced grid of 1,331 points over the coded cube, 79.9% of the characterized region satisfies all three criteria at once.

Aggregate is the response that rejects the remainder. It alone rejects 20.1% of the grid, XMuLV rejects 0.1%, and acidic charge variants reject none. The binding constraint on the operating region is therefore not the attribute the step sets. It is the attribute the step adds to and the polishing step removes.

The two constraints bound the hold from opposite sides, which is what gives the region its shape. At the high-pH, short-hold, low-temperature corner of the characterized region the model predicts an XMuLV log reduction of 4.91 $\log_{10}$ against the required step contribution of 4.93 $\log_{10}$, so that corner lies outside the design space and an edge of failure exists inside the knowledge space. At the opposite corner, with the pH, the hold and the temperature all at their high edges, the model predicts 2.382 % aggregate against an in-process criterion of 2.17 %, so that corner lies outside the design space as well. The principal planes of the region are shown in Figure 5.1, where the two gradients run along different axes.

The normal operating ranges lie well inside the region. At the worst corner of the normal operating ranges for virus inactivation the model predicts 6.43 $\log_{10}$, which clears the required step contribution by 1.50 $\log_{10}$. At the worst corner of the normal operating ranges for aggregate it predicts 2.079 %, inside the in-process criterion. The nesting is the one the enhanced approach describes. The normal operating ranges sit inside the design space, the design space sits inside the characterized region, and the characterized region is bounded by what was studied (International Council for Harmonisation 2009; CMC Biotech Working Group 2009).

The design space is bounded in three ways. It was established on a scale-down model and a small-scale spiking model, and it is confirmed at commercial scale only through the Stage 2 programme. It rests on models of mean response, so a setting on the boundary carries about half the reliability of a setting inside it, and the proven acceptable ranges in §7 are the form in which the variability is carried. It covers the three parameters studied multivariately, and A-Mab concentration enters it only through the range in which that parameter was assessed.

Movement within the design space is not a change from a regulatory filing perspective (International Council for Harmonisation 2009). It remains subject to the change management system of the pharmaceutical quality system, which assesses a planned move prospectively against existing product and process knowledge (International

Council for Harmonisation 2008). Movement outside the design space requires a new assessment and, where the move affects the clearance claim, new clearance data.

7 Proven acceptable ranges

A proven acceptable range is the range of one parameter over which one attribute stays inside its criterion. Table 7.1 gives the ranges for every combination of a governed attribute and a multivariate parameter, together with the characterization range each was computed over and the criterion each was judged against.

Table 7.1: Proven acceptable ranges by quality attribute and parameter.

CQA	Parameter	Char. range	Unit	Crite-rion	PAR (set-point)	PAR (NOR)
Aggregates (HMW, %)	Inactiva-tion pH	3.2–4	pH	≤ 2.17	3.2–4	3.2–4
Aggregates (HMW, %)	Hold time	60–180	min	≤ 2.17	60–161	60–144
Aggregates (HMW, %)	Tempera-ture	15–25	°C	≤ 2.17	15–25	15–25
XMuLV LRF ($\log_{10}$)	Inactiva-tion pH	3.2–4	pH	≥ 4.93	3.2–4	3.2–3.96
XMuLV LRF ($\log_{10}$)	Hold time	60–180	min	≥ 4.93	60–180	60–180
XMuLV LRF ($\log_{10}$)	Tempera-ture	15–25	°C	≥ 4.93	15–25	15–25
Acidic variants	Inactiva-tion pH	3.2–4	pH	≤ 40	3.2–4	3.2–4
Acidic variants	Hold time	60–180	min	≤ 40	60–180	60–180
Acidic variants	Tempera-ture	15–25	°C	≤ 40	15–25	15–25

The criteria in the table differ in origin, and the difference is stated here so that no range is read against the wrong yardstick. Aggregate was judged against the in-process criterion of 2.17 % described in §2.2, which is the level this step may hand on given the clearance the downstream train delivers. XMuLV was judged against the step contribution of 4.93 $\log_{10}$ that remains once the clearance credited to anion exchange and virus filtration is subtracted from the cumulative requirement. Acidic charge variants were judged against the drug substance criterion, which the step does not approach.

Two analyses were run for every combination. The first evaluates the fitted model along the parameter of interest with the other two parameters at fixed settings. The second varies the other two parameters inside their normal operating ranges by Monte-Carlo simulation of the fitted model, 2,000 draws at each of 81 settings, and adds the model residual to each draw. The second analysis reports the range over which the 95 % predictive interval of the attribute stays inside the criterion. It therefore answers a question about robustness, and it returns the narrower range wherever the two differ.

7 of the 9 combinations return the full characterization range under both analyses. In those cases no setting of the parameter inside its characterized range takes the attribute outside its criterion, with the rest of the operating point anywhere inside its normal operating ranges. The remaining 2 return less, and they are the combinations that constrain the step.

Hold time against aggregate is the tightest. The first analysis returns 60 to 161 minutes and the second 60 to 144 minutes, against a characterization range that runs to 180 minutes. The upper edge of the normal operating range is 120 minutes, which leaves 24 minutes of margin under the robustness analysis. Figure 7.1 plots the response against hold time under both analyses, with the acceptable region shaded.

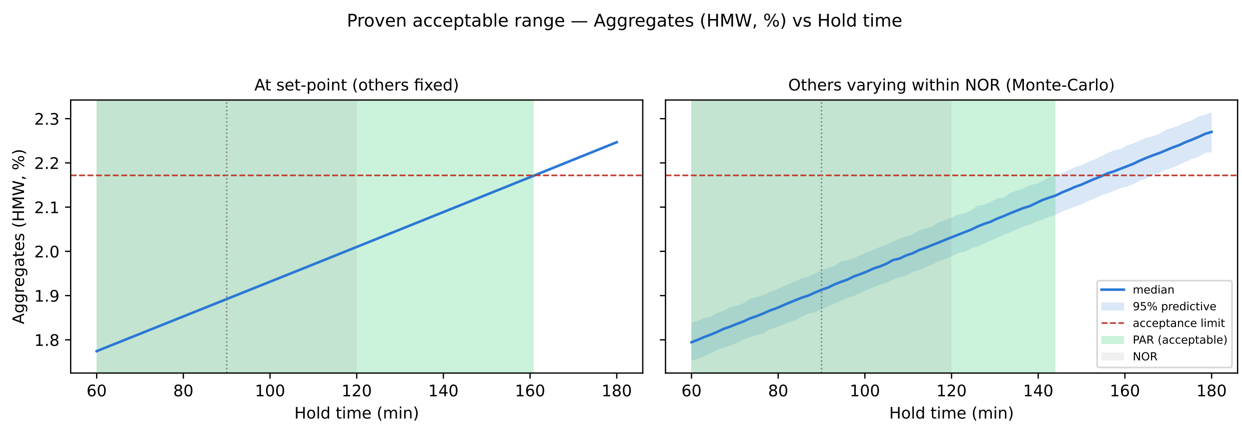

Figure 7.1: Proven acceptable range of hold time for aggregate, at fixed conditions (left) and with the other parameters varying inside their normal operating ranges (right).

Inactivation pH against XMuLV clearance is the second. The first analysis returns the full characterization range and the second returns 3.20 to 3.96, so the high edge of the characterized pH range is not robust to the rest of the operating point moving inside its normal ranges. The upper edge of the normal operating range for pH is 3.6, well below that limit. Figure 7.2 plots the clearance against pH under both analyses.

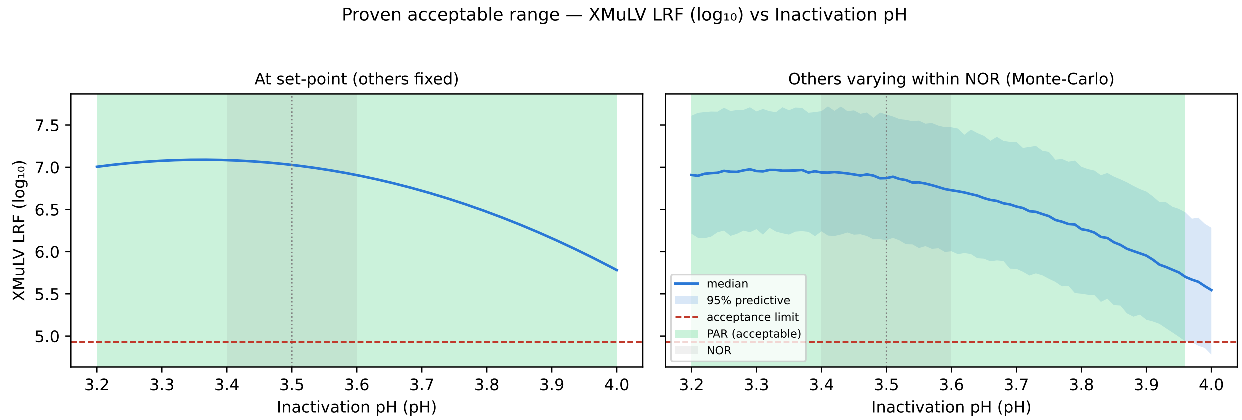

Figure 7.2: Proven acceptable range of inactivation pH for XMuLV log reduction, at fixed conditions (left) and with the other parameters varying inside their normal operating ranges (right).

The third governed attribute is plotted in Figure 7.3 on the same basis. Hold time against acidic charge variants returns the full range under both analyses. The response rises linearly with hold time and reaches 20.89 % at the longest hold studied, against a criterion of 40 %. The step does not constrain this attribute anywhere in the region studied.

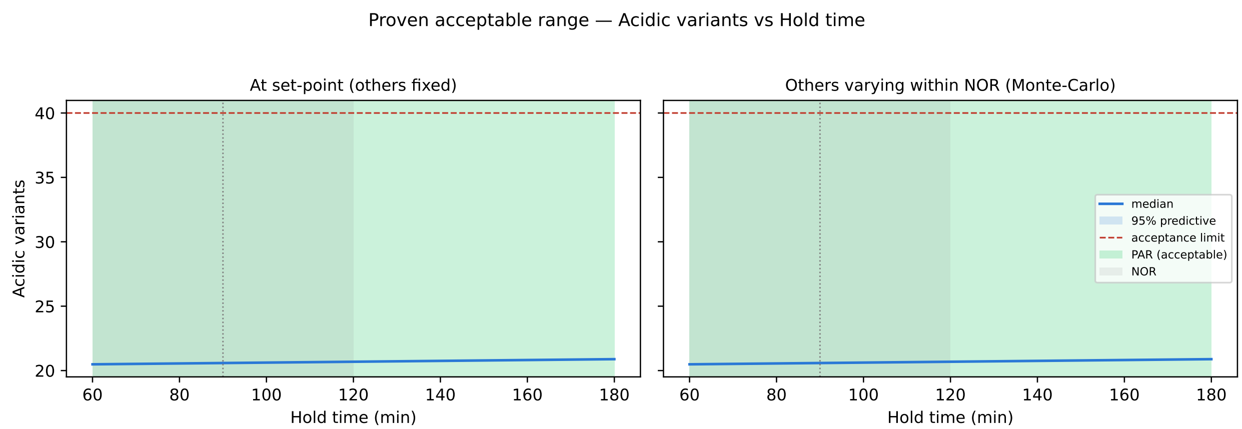

Figure 7.3: Proven acceptable range of hold time for acidic charge variants, at fixed conditions (left) and with the other parameters varying inside their normal operating ranges (right).

The ranges in Table 7.1 are univariate statements and the design space in §6 is a multivariate one, so the two do not substitute for each other. A proven acceptable range answers what one parameter may do while the rest of the operating point is held or varied as described. The design space answers where the parameters may go together. The 79.9% of the characterized region that satisfies every criterion in §6 is smaller than the product of the univariate ranges in Table 7.1, because a corner can fail on a combination that no single range excludes.

8 Process capability and robustness

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches. Every parameter of the train was drawn inside its normal operating range, the fitted models propagated those settings to the attributes, and the resulting distribution was compared with the acceptance criterion of each attribute. Table 8.1 gives the outcome for the three attributes this step governs. The values are drug substance values, so they carry the contribution of every step that acts on the attribute and not this step alone.

Table 8.1: Commercial-scale capability for the attributes this step governs.

CQA	Crit.	Spec	Acceptance	Mean ± SD	Cpk
Aggregates (HMW)	H	upper	0–5	0.753 ± 0.088	16.1
Acidic charge variants (deamidation)	VL	upper	18–40	22.3 ± 1.4	4.26
Viral clearance — XMuLV (enveloped)	VH	lower	16.7–30	19.5 ± 0.61	1.53

Clearance of XMuLV is the tightest of the three, at a one-sided Cpk of 1.53. The simulated mean sits 2.81 $\log_{10}$ above the cumulative requirement, and the standard deviation of 0.61 $\log_{10}$ is what turns that margin into the reported figure. Aggregate returns a Cpk of 16.1 and acidic charge variants 4.26, both of them against criteria the drug substance meets by a wide margin after the polishing steps have acted.

Capability against a viral safety attribute needs one qualification that the other two do not. The distribution simulated for XMuLV is the sum of the log reductions of the three steps that clear it, so the Cpk in Table 8.1 describes the train. This step contributes 7.10 $\log_{10}$ of the nominal 18.87 $\log_{10}$ total, and the remaining 11.77 $\log_{10}$ comes from anion exchange (PCR-008) and small-virus retentive filtration (PCR-009). The step is the largest single contributor, and it is not the whole claim.

Robustness inside the normal operating ranges follows from §6 and §7. No corner of the normal operating ranges takes a governed attribute outside its criterion, and the two parameters whose proven acceptable ranges are narrower than their characterization ranges retain margin to the edge of the normal operating range in both cases. Aggregate is the attribute with the least margin in relative terms, and it is also the attribute a downstream step is designed to remove.

The tightest capability in the drug substance is not reported here. It belongs to an attribute another unit operation sets, at a Cpk of 1.51, and it is consolidated with the rest of the campaign in PCMR-001.

9 Parameter classification

Parameters were classified under SOP-4001 from the effects demonstrated in §5 together with the risk of operating outside the acceptable region, which follows the continuum of criticality of ICH Q9 and the classification logic of the A-Mab case study (International Council for Harmonisation 2023b; CMC Biotech Working Group 2009). A parameter whose variability affects a critical quality attribute is critical. Whether it is a critical process parameter or a well controlled critical process parameter depends on how reliably it is held inside the range that assures the attribute.

- Inactivation pH was classified as a critical process parameter. It has the largest effect on virus inactivation of any parameter studied, at -2.08 $\log_{10}$ across the characterization range, and its proven acceptable range under the robustness analysis stops at 3.96, inside the range characterized. The parameter is set by a titration to an endpoint, which is a less reliable operation than holding a timer or a jacket temperature, so the risk of leaving the assured range is not negligible. Both grounds point the same way, and the parameter is controlled to the narrowest normal operating range of the step.
- Hold time was classified as a well controlled critical process parameter. It affects two critical quality attributes in opposite directions, raising clearance by +1.72 $\log_{10}$ and aggregate by +0.471 % across the characterization range, so it is quality-linked and cannot be a key process parameter. It is timed by the control system from a recorded start, which makes it one of the most reliably held parameters in the train, and its proven acceptable range leaves margin above the normal operating range in both directions.
- Temperature was classified as a well controlled critical process parameter. It acts on the same two attributes as hold time and in the same directions, at +0.64 $\log_{10}$ and +0.221 % across the characterization range, which are the smallest quality-linked effects of the three multivariate parameters. It is held by a jacketed vessel under closed-loop control and its proven acceptable range covers the whole characterization range for every governed attribute.
- A-Mab concentration was classified as a general process parameter. It does not enter the acid chemistry that inactivates the virus, and across the range assessed in §4.4 it did not move a governed attribute enough to warrant a control beyond the Protein A pool specification that already bounds it. The classification rests on the range assessed, and a change to the capture step that moved the pool concentration outside that range would require it to be revisited.

1 parameter of the step is a critical process parameter and 2 are well controlled critical process parameters, so 3 of the 4 parameters are quality-linked. None of the parameters was classified as a key process parameter, because each of the three

multivariate parameters acts on a critical quality attribute and the fourth acts on none. The counts for the whole train are consolidated in PCMR-001.

10 Contribution to the control strategy

The step contributes four elements to the control strategy. The first is the set of parameter ranges in Table 4.1, which are the ranges Stage 2 will confirm. Inactivation pH is controlled to the narrowest of them and is verified before the hold is started, so that the hold is timed from a confirmed condition. Hold time and temperature are controlled by the automation and recorded continuously.

The second is in-process monitoring of the attributes the step moves. Aggregate is measured on the neutralised pool against the in-process criterion of 2.17 % used throughout this report, which is the level cation exchange is designed to receive. The nominal at-scale output of the step is 1.91 % aggregate, of which 0.35 % is added by the hold itself. Charge variants are monitored at this step and are not controlled at it, since the change over the hold is small against the level the production bioreactor sets.

The third is the clearance claim itself. Table 10.1 gives the log reduction credited to each step for each model virus and the cumulative totals.

Table 10.1: Credited log reduction by step and model virus.

step	XMuLV	MVM
Low-pH Viral Inactivation	7.1	0
Anion Exchange (AEX)	5.95	4.71
Virus Filtration	5.82	5.32
Cumulative	18.9	10

This step is credited with 7.10 $\log_{10}$ against XMuLV and with no clearance of MVM, which is correct for the mechanism. A non-enveloped parvovirus has no lipid envelope for acid to disrupt, so a low-pH hold does not inactivate it, and the MVM claim rests entirely on anion exchange and virus filtration. Clearance of XMuLV by this step is inactivation rather than removal, and clearance by the other two steps is removal by charge and by size, so the three modules act by different mechanisms and their log reductions are added under ICH Q5A(R2) (International Council for Harmonisation 2023a). The value credited here is a minimum, for the reasons given in §3.3 and §5.4.

The fourth is the set of controls this step relies on and does not provide. The acid and base solutions are controlled by the buffer preparation procedure. The Protein A pool concentration is controlled by the capture step (PCR-005) and is the input that keeps A-Mab concentration inside the range assessed in §4.4. Aggregate formed during the hold is removed by cation exchange (PCR-007), and the in-process criterion used here is derived from that step's capability, so a change to the cation exchange operating region changes the criterion this step is judged against.

Life-cycle controls follow from the same dependencies. The scale-down model and the small-scale spiking model are requalified when the equipment train or the load material changes materially, the infectivity assay is maintained under AMV-3017, and the clearance claim is reassessed if the pH control range, the minimum hold time or the load composition changes. The step contributes no release test of its own. Its outcome is verified through the drug substance tests for aggregate and charge variants and through the modular clearance claim consolidated in PCMR-001.

11 Discussion

The step is now described by two predictive models over ranges wider than it will be operated in. What was known before the study was the mechanism and the direction of every effect, from three prior products on the same platform. What the study adds is the size of each effect for A-Mab, the shape of the surfaces, and the location of the boundary at which the region stops being acceptable. The most useful result is the separation between the parameter that governs clearance and the parameters that govern aggregate. Inactivation pH moves the clearance without moving the aggregate, so the two constraints can be satisfied independently over most of the region, and they meet only along the hold-time axis.

The second result worth stating is that the binding constraint is not the attribute the step sets. Aggregate rejects 20.1% of the characterized region and XMuLV rejects 0.1%, so the region is bounded by what the step hands to the polishing step. That is a consequence of the in-process criterion used in §2.2. A criterion drawn from the drug substance specification instead would have credited the hold with clearance that cation exchange performs, and would have reported the whole characterized region as acceptable.

Confidence in the scale-down model rests on two arguments of different strength. For aggregate and charge variants the model was compared with at-scale data on the same attributes, and the parameters that drive them are scale-independent, so the comparison is direct. For virus clearance no at-scale comparison is possible, because live virus may not be introduced into a manufacturing suite. The clearance claim therefore rests on the qualification of the spiking model against the process conditions, which is the position ICH Q5A(R2) takes for every modular clearance claim (International Council for Harmonisation 2023a).

Three limitations bound the results. The first is the scale-down model itself, and the design space is not confirmed at scale at the edges of the ranges. The second is the precision of the XMuLV response. Its predicted R^2 of 0.579 is well below its R^2 of 0.896, and the centre-point replicates vary by 0.38 $\log_{10}$, so the model predicts mean clearance better than it predicts a single run. This is managed by claiming the clearance as a minimum and by holding the normal operating ranges far inside the acceptable region. The third is the acidic charge-variant response, which returned no measurable variation between replicates and is therefore reported as a bound.

The two deviations recorded during execution are described in §13. Neither changed a conclusion. The hold-time record discrepancy was bounded against the fitted models and is smaller than the reproducibility of every response it could affect. The assay cytotoxicity raised the reportable floor of the infectivity assay for the affected samples, which makes the measured log reduction a lower bound and the clearance claim conservative. Both were retained, and the affected data are included in the analysis reported here.

12 Conclusions

The low-pH viral inactivation step has been characterized over ranges wider than its control ranges, on a qualified scale-down model and a qualified small-scale spiking model, against the protocol PCP-006.

The operating region is defined in §6. 79.9% of the characterized region satisfies every criterion the step governs, aggregate is the binding constraint, and every corner of the normal operating ranges lies inside the region with margin. The proven acceptable ranges in §7 justify each parameter range against each governed attribute, and 7 of the 9 parameter and attribute combinations return the full characterized range under both analyses.

The attributes meet their acceptance criteria at commercial scale. Clearance of XMuLV is the tightest of the three at a Cpk of 1.53, with the simulated mean 2.81 $\log_{10}$ above the cumulative requirement, and that figure is a property of the three clearance steps together and not of this step alone.

3 of the 4 parameters are quality-linked, with inactivation pH the single critical process parameter. 2 deviations were recorded, investigated, bounded and retained. The classification, the ranges and the clearance contribution reported here roll up into PCMR-001, and the ranges are confirmed at commercial scale in Stage 2.

13 Deviations from the plan

2 deviations from PCP-006 were recorded during execution. Both were raised in the quality system, investigated to root cause and dispositioned before the analysis was finalised, and both were retained with the affected data included. Table 13.1 is the register.

Table 13.1: Deviations recorded during execution of PCP-006.

Deviation	Summary	Detected during	Disposition
DEV-006-01	Hold-time record discrepancy between DCS and manual log	batch-record reconciliation	retained
DEV-006-02	XMuLV infectivity assay cytotoxicity at low dilution	assay review	retained

13.1 DEV-006-01 — Hold-time record discrepancy

During batch-record reconciliation the hold duration recorded by the distributed control system was found to differ from the duration entered in the manual log by 4 minutes. The discrepancy was present on the affected runs in the same direction and of the same size, which pointed at a systematic cause.

The investigation compared the two time sources against a reference clock and found an offset between the distributed control system clock and the clock used for the manual log. No evidence was found that any hold was executed for a duration outside the interval the two records bracket. The root cause was recorded as a clock synchronisation offset, and the records were corrected to the control system time, which is the timed source the hold is controlled from.

The impact was assessed by evaluating the fitted response-surface models at the set-point and at the set-point displaced by the recorded discrepancy. The predicted change is 0.0158 % aggregate, 0.024 $\log_{10}$ of XMuLV clearance and 0.013 % acidic charge variants. Each of those is at or below the standard deviation of the centre-point replicates for the same response (Table 5.1), so the offset is not distinguishable from the reproducibility of the step. A common offset also displaces every run in the same direction, which shifts the intercept of a fitted model without changing the estimated effects. The deviation was dispositioned as retained, and the affected runs were included in the analysis. The corrective action was to bring the manual log clock under the same synchronisation as the control system.

13.2 DEV-006-02 — Infectivity assay cytotoxicity at low dilution

Review of the XMuLV infectivity assay found cytotoxicity in the indicator cell monolayers at dilutions below one in 8 of the sample matrix. Cytotoxic wells cannot be scored for viral cytopathic effect, so the affected dilutions were excluded and the titre of the held pool was read from the first scorable dilution.

The investigation attributed the cytotoxicity to the sample matrix and not to the method. The neutralised hold pool carries the acid and base used in the titration and the antibody at process concentration, and cytotoxicity was observed only in the samples from the hold and not in the spiked load samples, which are diluted further. The method itself performed inside its validated acceptance criteria on the assay controls, and no repeat qualification of AMV-3017 was required.

Excluding the lowest dilutions raises the lowest titre the assay can report for the affected samples. A log reduction factor calculated from a titre at that floor is a lower bound on the reduction achieved, because the true titre may lie anywhere below the reported value. The consequence for this report is one-sided. Every affected log reduction factor under-reports the clearance, so the models fitted to those values, the criterion in §7 and the credited clearance in §10 are conservative. The deviation was dispositioned as retained. The corrective action was to add a matrix dilution step to the sample preparation for future clearance studies at this step.

14 Appendix A — Screening design matrix and responses

The executed screening design is given in coded form in Table 14.1 and in natural units in Table 14.2. The coded levels −1, 0 and +1 are the low edge, the midpoint and the high edge of each characterization range in Table 4.1.

Table 14.1: Screening design in coded factors, with the measured responses.

Run	Type	A	B	C	Aggregates (HMW, %)	XMuLV LRF ($\log_{10}$)	Acidic variants
1	facto-rial	-1	-1	-1	1.69	7.02	20.5
2	facto-rial	-1	-1	1	1.9	7.48	20.5
3	facto-rial	-1	1	-1	2.16	8.17	20.9
4	facto-rial	-1	1	1	2.32	9.37	20.9
5	facto-rial	1	-1	-1	1.69	4.81	20.5
6	facto-rial	1	-1	1	1.9	5.14	20.5
7	facto-rial	1	1	-1	2.14	6.62	20.9
8	facto-rial	1	1	1	2.44	7.17	20.9
9	center	0	0	0	2.01	7.35	20.7
10	center	0	0	0	2.07	6.92	20.7
11	center	0	0	0	2.01	7.36	20.7

Table 14.2: Screening design in natural units, with the measured responses.

Run	Type	Inactiva-tion pH	Hold time	Temper-ature	Aggregates (HMW, %)	XMuLV LRF ($\log_{10}$)	Acidic variants
1	facto-rial	3.2	60	15	1.69	7.02	20.5
2	facto-rial	3.2	60	25	1.9	7.48	20.5

Run	Type	Inactivation pH	Hold time	Temperature	Aggregates (HMW, %)	XMuLV LRF ($\log_{10}$)	Acidic variants
3	factorial	3.2	180	15	2.16	8.17	20.9
4	factorial	3.2	180	25	2.32	9.37	20.9
5	factorial	4	60	15	1.69	4.81	20.5
6	factorial	4	60	25	1.9	5.14	20.5
7	factorial	4	180	15	2.14	6.62	20.9
8	factorial	4	180	25	2.44	7.17	20.9
9	center	3.6	120	20	2.01	7.35	20.7
10	center	3.6	120	20	2.07	6.92	20.7
11	center	3.6	120	20	2.01	7.36	20.7

15 Appendix B — Response-surface design matrix and responses

The executed response-surface design is given in coded form in Table 15.1 and in natural units in Table 15.2. The axial runs sit on the faces of the coded cube, so every setting lies inside the characterization ranges in Table 4.1.

Table 15.1: Response-surface design in coded factors, with the measured responses.

Run	Type	A	B	C	Aggregates (HMW, %)	XMuLV LRF ($\log_{10}$)	Acidic variants
1	factorial	-1	-1	-1	1.69	6.49	20.5
2	factorial	-1	-1	1	1.88	7.35	20.5
3	factorial	-1	1	-1	2.17	7.65	20.9
4	factorial	-1	1	1	2.36	7.91	20.9
5	factorial	1	-1	-1	1.68	4.82	20.5
6	factorial	1	-1	1	1.88	5.77	20.5
7	factorial	1	1	-1	2.11	6.45	20.9
8	factorial	1	1	1	2.39	7.16	20.9
9	axial	-1	0	0	2.01	7.04	20.7
10	axial	1	0	0	2.02	6.13	20.7
11	axial	0	-1	0	1.79	6.65	20.5
12	axial	0	1	0	2.24	7.96	20.9
13	axial	0	0	-1	1.92	7.22	20.7
14	axial	0	0	1	2.13	7.29	20.7
15	center	0	0	0	2.02	6.94	20.7
16	center	0	0	0	1.98	6.15	20.7
17	center	0	0	0	2.02	6.97	20.7
18	center	0	0	0	2	6.78	20.7

Table 15.2: Response-surface design in natural units, with the measured responses.

Run	Type	Inactivation pH	Hold time	Temperature	Aggregates (HMW, %)	XMuLV LRF (log_{10})	Acidic variants
1	factorial	3.2	60	15	1.69	6.49	20.5
2	factorial	3.2	60	25	1.88	7.35	20.5
3	factorial	3.2	180	15	2.17	7.65	20.9
4	factorial	3.2	180	25	2.36	7.91	20.9
5	factorial	4	60	15	1.68	4.82	20.5
6	factorial	4	60	25	1.88	5.77	20.5
7	factorial	4	180	15	2.11	6.45	20.9
8	factorial	4	180	25	2.39	7.16	20.9
9	axial	3.2	120	20	2.01	7.04	20.7
10	axial	4	120	20	2.02	6.13	20.7
11	axial	3.6	60	20	1.79	6.65	20.5
12	axial	3.6	180	20	2.24	7.96	20.9
13	axial	3.6	120	15	1.92	7.22	20.7
14	axial	3.6	120	25	2.13	7.29	20.7
15	center	3.6	120	20	2.02	6.94	20.7
16	center	3.6	120	20	1.98	6.15	20.7
17	center	3.6	120	20	2.02	6.97	20.7
18	center	3.6	120	20	2	6.78	20.7

16 Appendix C — Full effect and coefficient tables

Table 16.1 gives the screening effect estimates for acidic charge variants, completing the set begun in §5.2. Table 16.2 gives the response-surface coefficients for the same response, which is the one response whose coefficients are not reproduced in §5.3. The standard errors in that table are at the numerical precision of the fit, for the reason given in §5.3.

Table 16.1: Screening effect estimates for acidic charge variants.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	0.4	0.2	4.12e-15	4.86e+13	1.08e-54	***
BC	-6.19e-15	-3.09e-15	4.12e-15	-0.751	0.494	
C	-5.35e-15	-2.67e-15	4.12e-15	-0.649	0.551	
AB	-4.57e-15	-2.29e-15	4.12e-15	-0.555	0.608	
AC	-2.6e-15	-1.3e-15	4.12e-15	-0.316	0.768	
A	6.69e-16	3.35e-16	4.12e-15	0.0812	0.939	

Table 16.2: Response-surface coefficients for acidic charge variants.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	20.7	3.39e-15	6.11e+15	5.79e-124	***
A	2.63e-15	2.72e-15	0.964	0.363	
B	0.2	2.72e-15	7.34e+13	1.32e-108	***
C	2.28e-15	2.72e-15	0.838	0.426	
AB	-1.19e-15	3.04e-15	-0.391	0.706	
AC	-2.19e-15	3.04e-15	-0.719	0.493	
BC	-3.12e-15	3.04e-15	-1.02	0.335	
A^2	-4e-15	5.23e-15	-0.764	0.467	
B^2	2.22e-15	5.23e-15	0.424	0.682	
C^2	-1.78e-15	5.23e-15	-0.34	0.743	

Table 16.3 gives the variance partition for the same response. The pure-error sum of squares is zero, so the lack-of-fit F cannot be formed.

Table 16.3: Analysis of variance for acidic charge variants.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	0.4	9	0.0444	5.99e+26	2.43e-106
Residual	5.93e-28	8	7.42e-29	nan	nan
Lack of fit	5.93e-28	5	1.19e-28	nan	nan
Pure error	0	3	0	nan	nan

17 Appendix D — Analytical methods summary

Table 17.1 lists the validated methods that produced the data in this report, with the performance reported by the method validations that are held for this step. The infectivity assay is the method the clearance claim rests on, and its quantitation limit is what makes a log reduction measured against a titre at that limit a lower bound (§3.3 and §13.2).

Table 17.1: Reported performance of the validated methods held for this step.

Reference	Method	Precision (%)	LOQ	LOQ unit	Accuracy (%)
AMV-3017	XMuLV Infectivity Titre (TCID50)	0.3	1.5	log10 TCID50/mL	96

The size-variant and charge-variant methods used at this step are the corpus methods AMV-3011 and AMV-3013 listed in Table 3.1. Their validation summaries are held with the method validation reports and are not reproduced here.

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